

CHARLES AWUAH COBBINAH

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PROFILE

MSc Automotive Engineering student specializing in Functional Safety, Vehicle Systems Architecture, and Dynamic Simulation. Hands-on industrial experience in hazard analysis, FMEA, Fault Tree Analysis, machine safety, and autonomous vehicle operational risk assessment. Currently contributing to two tunnel vehicle safety engineering programs: a standard machine risk analysis platform and a driverless autonomous transport vehicle project focused on operational modes, sensor-based safety, system architecture, and risk mitigation. Proficient in MATLAB/Simulink and AMESim modeling. Seeking an apprenticeship in Automotive Safety Engineering.

EDUCATION

2025 – PRESENT | MSc Automotive Engineering for Sustainable Mobility M2| Institut Supérieur De L'Automobile Et Des Transports (ISAT) - Nevers, France

2024- 2025 | MSc Automotive Engineering for Sustainable Mobility M1| Polytech d'Orleans - Orléans, France

2019-2023 | B. TECH Automobile Engineering | Takoradi Technical University| Takoradi - Ghana

TECHNICAL PROJECT

Standard Tunnel Vehicle Safety Analysis Platform

- Developed a digital risk analysis platform for conventional underground transport machines.
- Structured FMEA workflows across braking, steering, fire, propulsion, and tilt monitoring systems.
- Implemented RPN prioritisation across 200+ failure modes, risk heatmaps, and automated engineering reporting for machines.
- Defined 24 safety functions with Performance Level (PLr) assignments, covering personnel detection, obstacle avoidance, zone speed enforcement, and emergency stop.

Driverless Tunnel Vehicle Functional Safety Project (with CEA collaboration) in Progress

- Performing complete hazard analysis for autonomous underground transport vehicles operating in manual, teleoperation, and fully autonomous modes.
- Applied ISO 26262 HARA methodology to determine ASIL levels for electric propulsion safety functions.
- Defined operational zones, mode-dependent hazards, and safety constraints.
- Developed safety concepts for obstacle detection, emergency stop, speed limitation, positioning integrity, and communication loss response.

Tire Vehicle Dynamic & ABS Conventional Braking Systems | MATLAB/Simulink |AMESim

- Designed and implemented a sliding mode ABS controller, creating a simulation process to validate braking performance and predict thermal fading on low-friction surfaces.
- Analyzed energy consumption on a roller test bench and distribution of hybrid gasoline Internal Combustion Engine using AMESim to identify key loss areas.

PROFESSIONAL EXPERIENCE

FEB 2026| MACHINE RISK ANALYSIS SAFETY INTERN. (BUREAU D'ÉTUDES)| METALLIANCE – Corail Tunnelier| Saint Vallier, FR
Internship Subject: Risk Analysis and Definition of a Secure Access Zone for an Autonomous Machine

- Conducted hazard identification and risk assessment for autonomous machine operation across multiple operating modes and restricted zones.
- Designed redundant architecture (LiDAR + Camera + Radar) achieving Category 3/4 compliance for PLd/PLe safety functions.

FEB 2024 – AUG 2024| MAINTENANCE RELIABILITY ENG. | QUANTUM TERMINALS LTD - Oil & Gas | Tema, Ghana

- Applied FMEA (Failure Mode and Effects Analysis) to proactively identify and mitigate critical process risks, including the precise control of butane/propane tank temperatures and the prevention of loading hose failures, ensuring safety.
- Supervised construction of a 5 km 12-inch steel hydrogen pipeline ensuring 100% compliance with ASME & EN standards, including the installation and integrity verification of a 3000 MT spherical storage tank.

APR 2022 – JAN 2024| FIELD SERVICE ENG. | BLACKCAD ENGINEERING AND LOGISTICS -GoldFields Mines| Tarkwa, Ghana

- Assembled and Validated engine performance and powertrain calibration post-rebuild through dynamometer testing and data analysis, ensuring compliance with performance specifications.

EXPERTISE & TECHNICAL SKILLS

ENGINEERING DOMAINS: Functional Safety | Vehicle Systems Architecture | Autonomous Systems Safety | Powertrain Systems | Component Failure Diagnostics.

METHODS & STANDARDS: Hazard Analysis | Risk Assessment | FMEA & AMDEC | Fault Tree Analysis (FTA) | PHA | HARA | ASIL/ PL| ISO 13849 |ISO 12100 | Requirements Analysis | Root Cause Analysis | Technical Documentation | Teamwork.

SOFTWARE SKILLS

MATLAB and Simulink| Capella| Python| Microsoft Office Suite | AMESim.

LANGUAGES:

English | C2 – Native || French | B2 – Intermediate